FALL SEMESTER 2026–2027
BCSE308P – COMPUTER NETWORKS
Innovative Evaluation Practices for Laboratory

REVIEW 1 – PROBLEM IDENTIFICATION & PLANNING

Drone-Based Real-Time Pandemic Monitoring for Emergency Search & Rescue

1. Project Title

Drone-Based Real-Time Pandemic Monitoring and Emergency Search & Rescue System using IoT and Wireless Networks

2. Introduction

Pandemics and mass-casualty emergencies demand rapid situational awareness across large, potentially inaccessible areas. Traditional ground-based monitoring is slow, risky for responders, and geographically limited. This project proposes an autonomous drone network that collects real-time biometric and environmental data, transmits it over wireless protocols, and aids rescue coordination

directly leveraging core Computer Networks concepts such as MQTT, mesh routing, and QoS-aware data delivery.

3. Problem Statement

During pandemics or disasters, health authorities and rescue teams lack real-time spatial data on :

Crowd density and hotspot zones

Individuals in distress (injured, unconscious)

Environmental hazard levels (temperature, air quality)

Manual monitoring is too slow, dangerous, and unscalable in such scenarios.

4. Motivation

COVID-19 exposed the critical gaps in large-area health surveillance

Search & rescue operations suffered from delayed response due to terrain barriers

Drone + IoT + CN protocols can bridge these gaps in a accost effective manner

Growing need for autonomous, networked emergency response systems

5. Objectives

Design a drone mesh network for real-time data relay

Implement live video + sensor data streaming over MQTT/WebSocket

Detect distress signals (heat signature, sound) using onboard sensors

Provide a centralized dashboard for emergency coordinators

Minimize latency in data transmission using QoS-aware routing

6. Existing System

- Manual ground patrols and static CCTV cameras
- Helicopter-based aerial surveys (expensive and slow)
- Contact tracing apps (no real-time spatial mapping)
- Isolated sensor networks without aerial mobility

7. Limitations of Existing System

Limitation	Impact
No real-time aerial view	Delayed emergency response
Human responder risk	Casualties among rescue teams
Static infrastructure	Cannot cover dynamic disaster zones
No sensor fusion	Incomplete situational picture

8. Proposed System

A fleet of GPS-enabled drones equipped with the following components:

- **Thermal + RGB cameras** — for victim detection
- **Air quality / temperature sensors** — for pandemic hotspot mapping
- **Wi-Fi / Zigbee mesh** — for inter-drone communication
- **4G/LTE uplink** — to a central command server
- **MQTT broker** — for lightweight real-time data publishing
- **Web dashboard** — for live map visualization

9. Novelty / Innovation

- **Drone mesh network** — drones act as mobile relay nodes, not just the data collectors
- **CN-layer focus** — QoS, routing the protocol adaptation (AODV/OLSR) for dynamic topology
- **Dual-mode operation** — pandemic surveillance + search & rescue in one system
- **Edge processing** on drone for latency reduction before uplink

10. Literature Survey

The following research areas form the foundation of this project:

- UAV-assisted IoT networks for disaster management
- MQTT performance in delay-tolerant and lossy networks
- FANET (Flying Ad-hoc Networks) routing protocols — AODV vs OLSR
- Thermal imaging for crowd health monitoring during pandemic scenarios
- QoS mechanisms in wireless mesh networks for emergency communications

References will be drawn from IEEE Xplore, Springer, and Google Scholar (5–8 papers).

11. Tools and Technologies

Layer	Technology
Hardware	Raspberry Pi / Arduino on drone
Sensors	DHT22, MQ-135, Thermal Camera
Network Protocol	MQTT, WebSocket, 802.11 Mesh
Routing	AODV / OLSR (FANET)
Backend	Python Flask / Node.js
Dashboard	React.js + Leaflet.js (Live Map)
Simulation	NS3 / GNS3 for CN Layer Testing

12. Initial System Architecture / Workflow

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[Drone Fleet] → sensors + camera
↓
[Drone Mesh Network] ←→ [Relay Drone]
↓ MQTT over Wi-Fi / LTE
[Cloud MQTT Broker]
↓
[Command Dashboard] → [Alert System]
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13. Project Timeline

Week 1-2

Literature Survery,finalizing architecture

Week 3-4

Hardware/simulation setup,protocol selection

Week 5-6

Network Layer implementation

Week 7-8

Sensor integration

Week 9-10

Dashboard Development

Week 11-12

Testing,performance analysis,Review 3 preparation.

14. Team Member Responsibilities

Shrinithish – Network Protocol design

Abhishek – Drone Sensor Integration

Shadhaan – Backend Server